

AI based integrated solution for smart pepper farming in Sri Lanka

Presented by 25-26J-479





Introduction to problem

- Pepper is one of the most valuable crops, but farmers face big challenges that reduce their income and stability.
- Prices often change unfairly, and there's no proper system to measure pepper quality
- Many new farmers also lack knowledge on how to detect pests and diseases early, leading to lower harvests.
- On top of that, most farmers can't predict how much yield they will get, making it hard to plan resources and manage their farms effectively.

These problems show why solutions are urgently needed to support farmers and strengthen the pepper industry.

Research Impact of the solution

Agricultural Productivity and Efficiency

- Moves farmers from guesswork to data driven decision. This can significantly increase yields, reduce losses, and optimize input usage.

Economic and Market Benefits

- AI based market price forecasting reduces the risks of selling at low prices, enabling farmers to make informed selling decisions.
- Post harvest AI and IoT improve consistency, which can help farmers capture higher export prices.
- Strengthens the international brand of Ceylon pepper by ensuring standardized quality.

Transparency and Trust in Supply Chains

- Blockchain enabled traceability enhances trust among exporters, buyers, and regulators

Previous research and its shortcomings

No AR based visualization techniques used

Most research are based on prediction of yield only by weight, no visualization techniques used for fruit yield visualization

Lack of a multiparameter grading system for pepper

There are no researches conducted on giving a grade based on images of pepper stock (mold, adulteration, size variation) and bulk density.

Lack of an integrated model for forecast market price

No integrated multivariate model for accurate forecasts and integration with AI forecasting for blockchain

No solution developed to detect spread percentage of diseases

Traditional disease detection in black pepper primarily focuses on classifying leaf diseases without estimating the extent of infection (disease spread percentage)

Our approach to address shortcomings



**Yield Prediction
& Visualization**



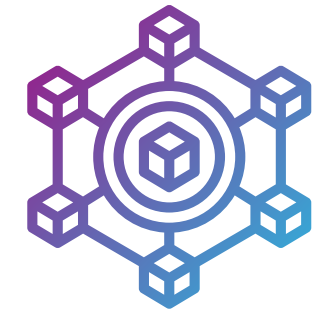
Disease Detection



**Post-Harvest
Quality Grading**



**Price
Forecasting**



**Blockchain
Integrity**

Constraints and Limitations

Pipeline content estimation

Only estimate piperine content based on variety metadata (lab-based testing is costly)

Mobile phone usage for image capturing

Farmers only use mobile cameras, model accuracy depends on consistent photo capture

Short-Term Forecasting Focus

The system is optimized for weekly forecasting and not for long-term market predictions.

Blockchain Constraints

The blockchain records data from system components (price and grade of quality), but does not cover entire logistics or financial transactions end-to-end.

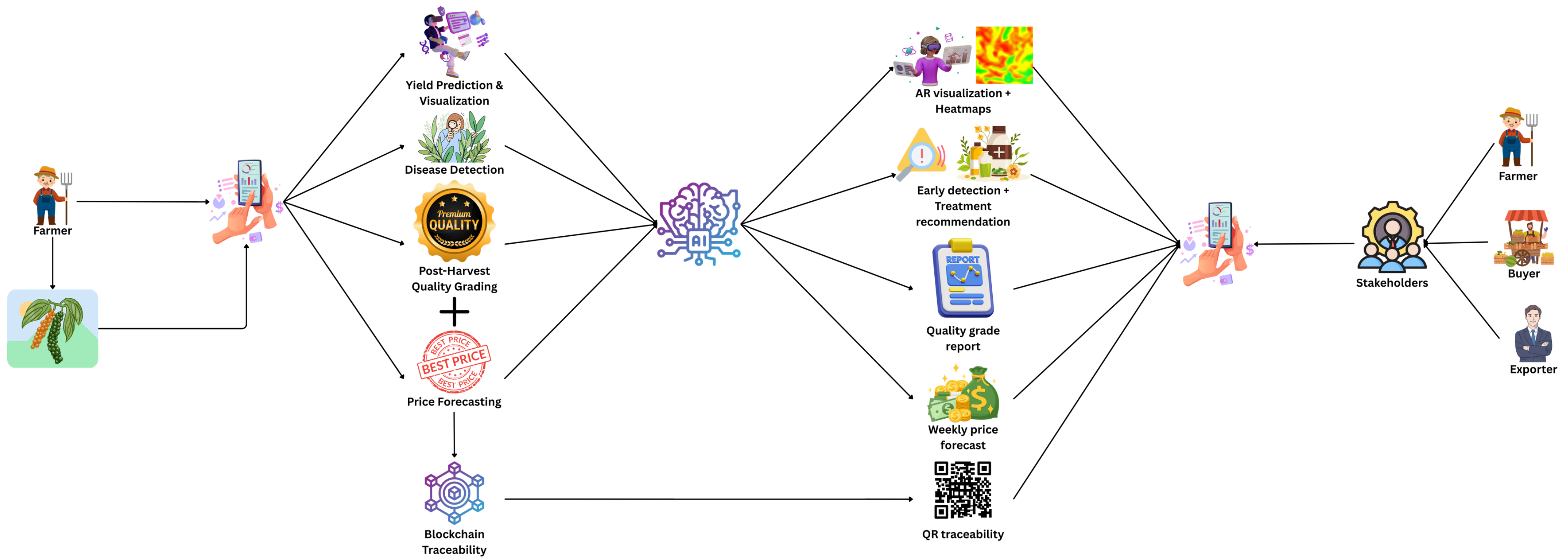
Model generalization

Model trained on Sri Lankan pepper varieties may not work well in other regions unless retrained.

Drone Usage in Heatmap Resolution

Without drone usage, heatmaps generation rely on farmer captured images via mobile phone.

SYSTEM DIAGRAM



Yield Prediction and Visualization

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Research Gap and Proposed Solution

Research Gap

- Yield prediction only by weight[1], yield prediction in pre planting and growth phase[4]
- AR used for maturity checking and grading only[1], AR/VR used for growth simulation of crop not fruit[2]

Proposed Solution

- Early stage yield prediction from young stages of pepper cone
- AR simulation of future growth of pepper cone in the plant
- Heatmap visualization of ripening/raw cone distribution

Knowledge Gap / Problem Definition

- Lack of integrated early stage yield prediction and visualization
- Unreliable forecasts at early growth stages make it difficult to allocate resources

Limitations of current systems and how the proposed solution overcomes them

Feature	ARIA[1]	Smart Industry Conference[2]	AR based Virtual Crop Planning & Yield Prediction [3]	Modeling and Visualization of Crops Growth Based on AR with Greenhouse Effect [4]	Khet-E system [5]	Proposed
Early stage yield prediction from young stages of fruit	✓ Ripe/unripe classification and grading in early stage and prediction of weight of yield	✗ Growth simulation under Green House Effect	✓ Yield prediction in pre planting and growth phase	✗ Growth modeling of crop under greenhouse effect	✗ Doesn't fully capture long-term crop dynamics	✓
AR simulation of future growth of fruit	✗ AR used for maturity checking and grading only	✗ AR/VR used for growth simulation of crop not fruit	✗ AR for 3D crop growth visualization	✗ AR for plant growth evaluation	✗ AR smart cards	✓
Heatmap visualization of ripening/raw cone distribution	✗	✗	✗ Limited(plantation is scanned via AR)	✗	✗ Locust attack prediction with map	✓

Key domains of knowledge

Artificial Intelligence & Machine Learning (Computer Vision, CNN)

Augmented Reality (ARCore/ARKit for interactive visualization)

Geospatial Data Analysis (heatmaps for plantation-level yield)

IoT and Embedded systems

Latest technologies to be applied

Mobile App Development



AI/ML frameworks for AI model development



AR visualization and heatmap generation



Backend /Data Management



How to Evaluate/Prove the Success

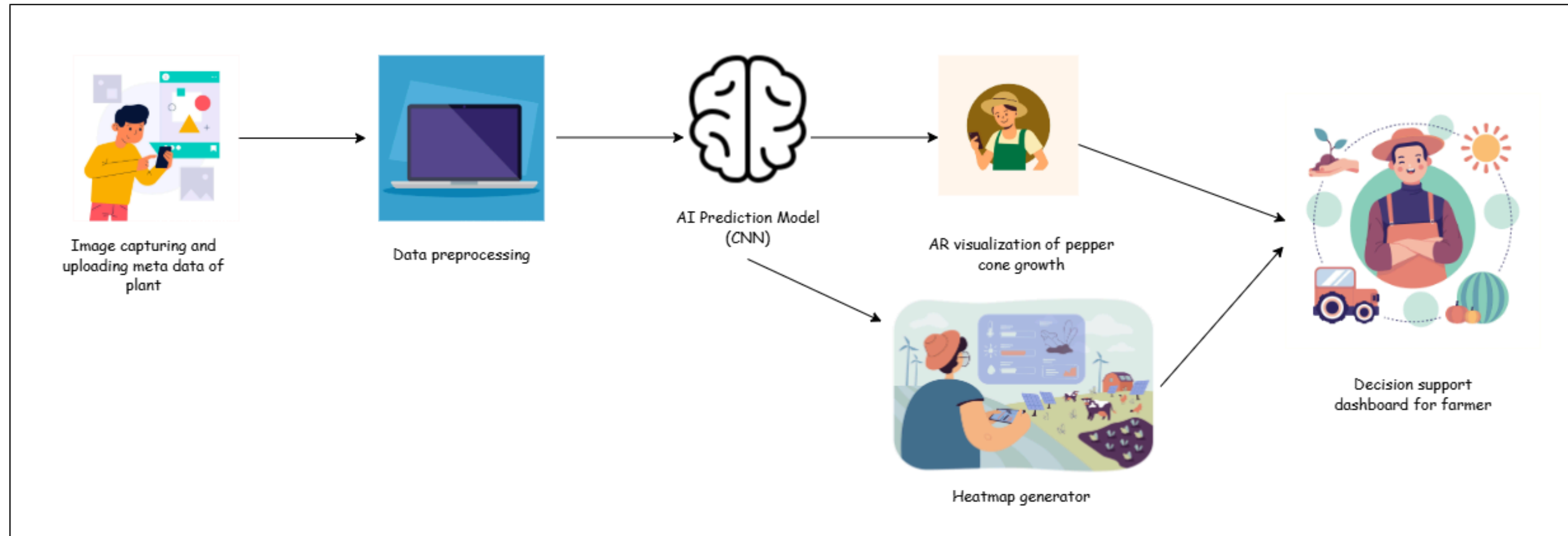
- Test with farmers and agriculture officers.
- Collect feedback through questionnaires
- Deploy system in pepper plantations for a season and track predicted vs actual yields and farmer decisions influenced by the app.
- AI Model Validation by expected accuracy >85%

Data availability and ethical clearance

Type of Data	Source	Parameters	Volume
Image data of pepper cone and plant	Field visits to Sri Lankan pepper plantations (selected Kegalle District)	Images of young pepper cones from multiple plants and images after growth of pepper Metadata with date, time, location, plantID, growth stage	Dataset of 800-1000 images from 200 pepper plants
Weather and Climate data	OpenWeatherMap API	Daily temperature Humidity Rainfall Sunlight hours	Collecting for 2-3 months per growing cycle
Soil Data	IoT sensors	Soil pH Moisture Nutrients levels	Collection from pepper plantations
Yield Data	Manual measurements at harvest	Cone counts and harvested weight per plant Cone count per plant Date of picking	200 plants across multiple plantations to ensure diversity.

- Farmers’ consent was obtained before capturing images or collecting yield data.
- Data will be anonymized (no farmer names, only plant IDs and GPS coordinates).
- Datasets will not be published without permission; instead, a research-only dataset will be created.

System Overview of the Individual Component



References

- [1] B. V. B. Prabhu, R. Shashank, B. Shreyas, and O. S. J. Narsipura, "ARIA: Augmented Reality and Artificial Intelligence enabled mobile application for Yield and grade prediction of tomato crops," in *Procedia Computer Science*, vol. 235, pp. 2693–2702, 2024.
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- [4] D. V. Goncharov, O. A. Ivashchuk, and N. G. Reznikov, "Method for Modeling and Visualization of Agricultural Crops Growth Based on Augmented Reality Technology in Terms of the Greenhouse Effect Dynamics," in *Proc. 2023 Int. Russian Smart Industry Conf. (SmartIndustryCon)*, Belgorod, Russia, 2023, pp. 66–71.
- [5] K. Kalaiselvi, J. J. Joseph, and G. Bosco, "Crop Yield and Locust Prediction using Machine Learning with Augmented Reality Smart Cards (Khet-E System)," *Indian Journal of Natural Sciences*, vol. 13, no. 71, pp. 473–480, Apr. 2022. ISSN: 0976–0997.

Post-Harvest Pepper Quality Grading



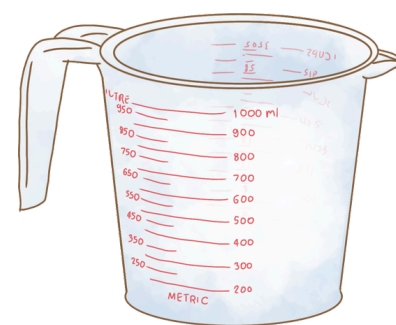
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Problem & Proposed Solution

Knowledge Gap

- Inaccurate Grading
- No Integrated System
- Lost Opportunities



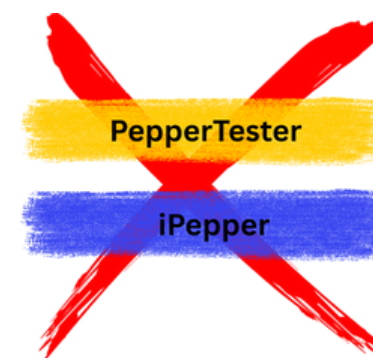
Creative Solution

- Multi-parameter grading system
 - IoT Device
 - AI Image Analysis
 - Certification
- Digital grading reports



Shortcomings of Existing Systems

- Traditional Cup Method → only measures bulk density, inaccurate & subjective
- ISO/IPC Standards [1-4] → comprehensive, but lab-only & not farmer-accessible.
- PepperTester [5] → checks seed purity only, not multi-parameter
- iPepper [6] → limited image + moisture data, not farmer-friendly



Comparison Between Existing Systems

Research / Method	Bulk Density	Visual Defects	Piperine (Chemical)	Certification	Multi-Parameter	Farmer-Friendly
Traditional 1L Cup Method	✓	✗	✗	✗	✗	✓ (simple, but inaccurate)
ISO 959 / IPC Standards [1 - 4]	✓ (required)	✓ (required)	✓ (lab only)	✓	✗	✗ (benchmarks, not tools)
PepperTester (2025) [5]	✗	✓ (purity only)	✗	✗	✗	✗
Piperine Studies [7 - 9]	✗	✗	✓ (lab-based only)	✗	✗	✗
iPepper (2011) [6]	✗	✓ (limited, image + moisture)	✗	✗	✗	✗
Proposed Research	✓ (IoT: bulk density)	✓ (AI: mold, color etc.)	✓ (variety-based estimation)	✓ (GAP certification)	✓ (fully integrated)	✓ (mobile app with reports and feedback)

Application of Specialization Knowledge & Technologies

Key Domains of Knowledge



IoT & Embedded Systems



Artificial Intelligence & Computer Vision



Software Engineering

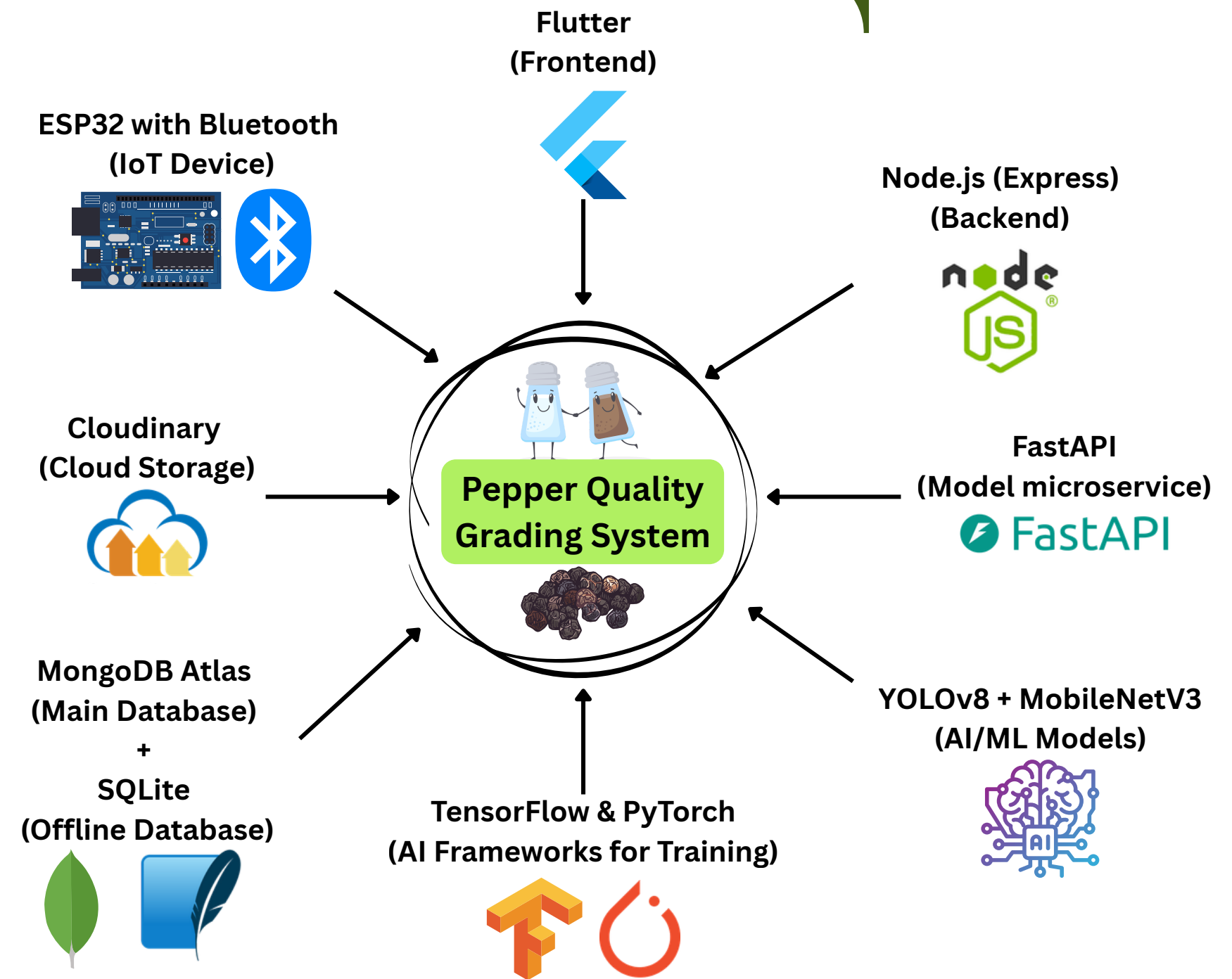


Databases & Cloud Computing



Agricultural Science & Quality Standards

Latest Technologies Utilized



Data Availability

Type of Data	Source	Parameters	Planned Volume
Pepper Images	Field visits to Sri Lankan pepper farms	Mold, adulteration, berry size, color variations	~800 - 1000
Bulk Density Standards	ISO/IPC standards, research papers [1–4]	Reference values	Literature-based
Piperine Data	Published chemical analysis studies [7-9]	Piperine % per variety	Literature-based
Certification Data	GAP Certification API (https://slgap.doa.gov.lk/certificates.php)	Certificate validity & details	API-driven checks

Ethical Considerations & Evaluation Plan

Ethical Considerations

Farmer Consent → Will be obtained before capturing data/images.

Anonymization → No personal identifiers, only plant IDs & grading data.

Data Security → Encrypted storage in MongoDB Atlas & Cloudinary.

Fair Use → Data only for research & farmer benefit, not resale.

Transparency → Farmers receive grading reports, ensuring trust.

Evaluation Plan

AI Model Validation → Test AI accuracy using pepper image datasets. Expected accuracy: >85% for defect detection (mold, adulteration, color).

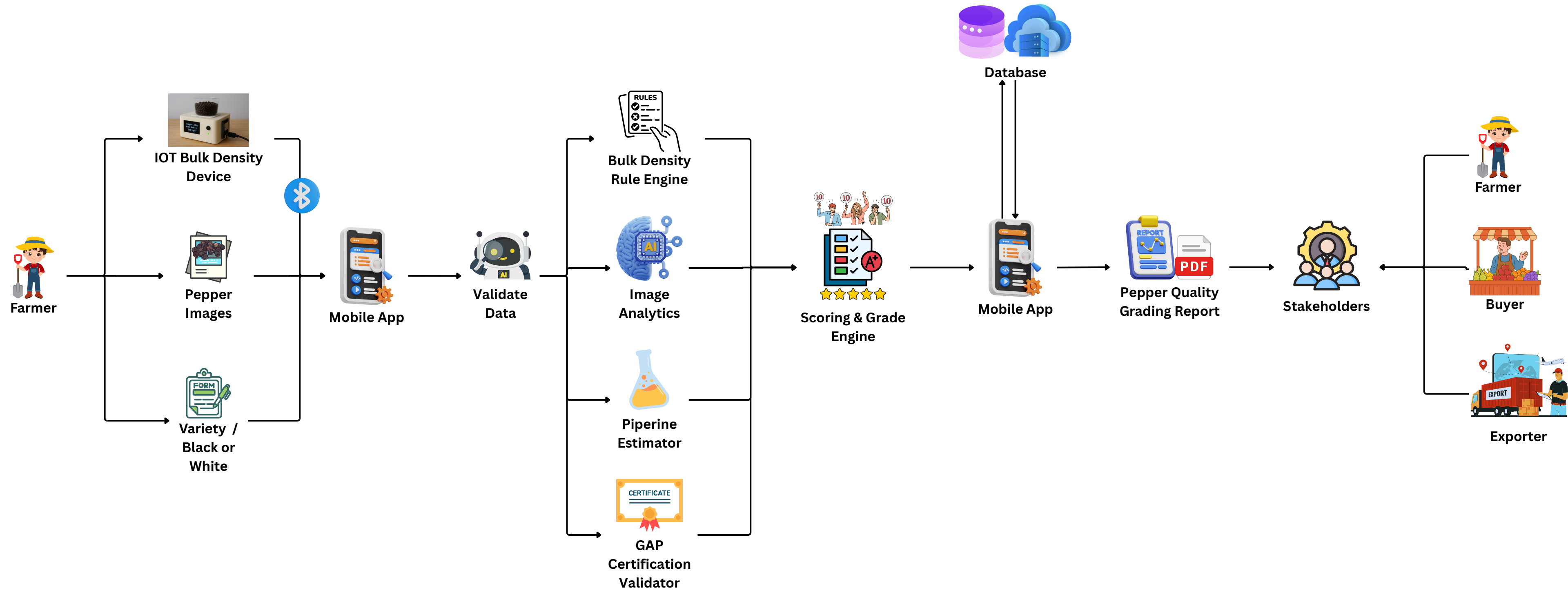
Benchmark Comparison → Compare bulk density & grading output against ISO/IPC standards and published piperine data.

Field Testing → Deploy app with selected farmers and agriculture officers for pilot testing.

User Feedback → Collect farmer & buyer feedback through surveys and focus groups.

Success Indicators → Higher grading accuracy, reduced export rejection, improved farmer trust.

System Overview



References

- [1] ISO, ISO 959-1:1998 – Pepper (*Piper nigrum* L.), whole or ground – Part 1: Black pepper, 1998.
- [2] ISO, ISO 959-2:1998 – Pepper (*Piper nigrum* L.), whole or ground – Part 2: White pepper, 1998.
- [3] Codex Alimentarius, Standard for Black, White and Green Peppers (CXS 326-2017), 2017 (rev.).
- [4] International Pepper Community (IPC), Standard Specifications for Pepper, 2023.
- [5] D. D. Minh, A. Author, and B. Author, “PepperTester: A non-destructive approach to assess the purity of black pepper seed samples,” [Journal/Conference Name], 2025.
- [6] D. N. F. Awang Iskandar, A. Author, and B. Author, “iPepper: Intelligent pepper grading and quality assurance system,” in Proc. IEEE CSPA, 2011, pp. xx–xx.
- [7] S. Author and T. Author, “Determination of piperine in pepper (*Piper nigrum*),” J. Chromatogr. A, vol. xx, no. xx, pp. xx–xx, 1983.
- [8] The Pharma Innovation Journal, “Evaluation of piperine content in black pepper,” Pharma Innov. J., vol. 12, no. 4, pp. 140–142, 2023.
- [9] H. Jayasinghe, A. Author, and B. Author, “Piper species (Piperaceae) of Sri Lanka and the diversity of the economically important *Piper nigrum*,” [Journal/Conference Name], 2020.

Market Price Prediction & Blockchain Integrity

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Research Gap and Proposed Solutions

Research Gap

- Traditional models ignore external factors (pepper type(black/white), grade of quality). [1]
- Advanced ML models are "black boxes" reducing trust. [2, 3]
- Blockchain faces adoption barriers in developing economies. [4]
- Forecasting and traceability are treated as separate systems. [5, 6]
- Lack of actionable insights for farmers from forecasts. [7]

Proposed Solutions

- Use Temporal Fusion Transformer (TFT) that integrates multivariate data for accurate forecasts.
- Integrate AI forecasting with blockchain to create a single, holistic solution.
- Provide clear "sell/wait" recommendations based on interpretable model outputs.

Knowledge Gap

- Current forecasting models are inaccurate and untrustworthy because they operate as "black boxes" ignoring critical external factors like quality grade and pepper type, and failing to provide farmers with clear, actionable advice.
- Currently, there is no any proper existng blockchain traceability systems for pepper industry in Sri Lanka.

Key Domains of Knowledge & Latest Technologies

TFT for Interpretable Time-Series Forecasting

- Multivariate price forecasting with explainable AI insights.
- Explainable and provide clear recommendations.

Blockchain Integrity

- Immutable record-keeping for supply chain transparency.
- Generates a QR Code, and provides the price details along with other details. (Grade)
- Practical and trustworthy, rather than complicated.

Evaluation & Success Metrics

- Validation with Historical Data – Compare predictions with actual past values to assess accuracy.
- Expected Accuracy – Aim for high prediction reliability (>85% for price forecasting).

Latest Technologies

- Flutter
- FastAPI, Node
- FireBase, Cloud Storage
- Python, Tensorflow, Jupyter

Data Availability and Ethical Clearance

Market Price Prediction

- **Historical Market Price Data:** From Department of Agriculture and Export Development Board
- **Weather Data:** From public API (OpenWeatherMap)
- **Grade of Quality:** From the Post-Harvest Quality Grading Component

Blockchain Integrity

Data from other components (Quality Grading and Market Price Prediction) will be used as inputs for the blockchain.

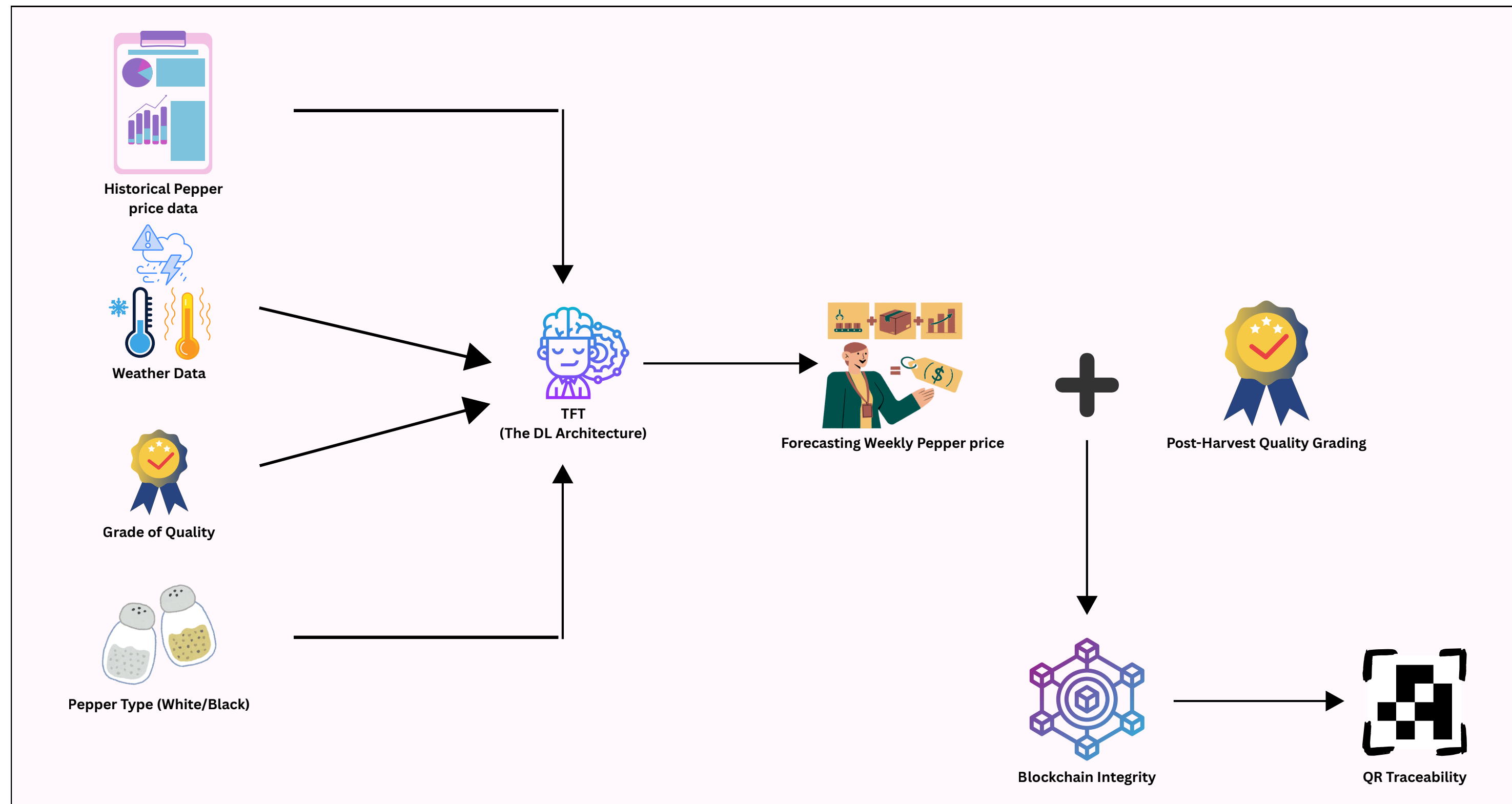
Ethical Clearance:

Privacy: Farmer personal data will be anonymized for model training. The blockchain will only record product and prediction data, not personal identifiers.

Transparency: Users will be informed via a clear privacy policy.

Access Control: The permissioned blockchain ensures that only authorized users can access sensitive data.

System Overview



References

- [1] L. Wang and Z. Li, "Limitations of ARIMA models in agricultural price forecasting under volatile market conditions," *Agric. Econ.*, vol. 68, no. 4, pp. 145–152, Apr. 2022.
- [2] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. KDD '16*, 2016, pp. 785–794.
- [3] F. Doshi-Velez and B. Kim, "Towards A Rigorous Science of Interpretable Machine Learning," *arXiv:1702.08608*, Feb. 2017.
- [4] V. S. S. Yadavalli and S. S. Sundar, "Barriers to Blockchain Adoption in Developing Economy Agri-Supply Chains: A Case Study of Indian Spices," *J. Agribus. Dev. Emerg. Econ.*, vol. 13, no. 2, pp. 245–262, 2023.
- [5] S. Kamble, A. Gunasekaran, and H. Gawankar, "Achieving sustainable performance in a data-driven agriculture supply chain: A review for research and applications," *Int. J. Prod. Econ.*, vol. 219, pp. 179–194, Jan. 2020.
- [6] S. J. Lee and J. U. Kim, "A Novel Framework for Agricultural Supply Chains: Integrating TFT Forecasting with Blockchain-Based Data Verification," *Electronics*, vol. 12, no. 4, p. 987, Feb. 2023.
- [7] Y. Han, Y. Liu, and L. Wang, "Economic system forecasting based on temporal fusion transformers: An interpretable approach," *Neural Netw.*, vol. 158, pp. 225–234, Feb. 2023.

AI powered disease detection system

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Research Gap and Proposed Solution

Research Gap

- Traditional disease detection in black pepper primarily focuses on classifying leaf diseases without estimating the extent of infection (disease spread percentage).
- Existing systems also often fail under real field conditions due to challenges like varying lighting, overlapping leaves, and partial visibility.

Proposed Solution

- Develop an AI-driven system combining image processing and CNN deep learning that not only classifies black pepper leaf diseases but also quantifies disease spread across leaves.

Knowledge Gap / Problem Definition

- Environmental influences (temperature, humidity, soil moisture) on disease progression are not adequately integrated into existing detection models.
- Field variability challenges (lighting differences, overlapping leaves, partial leaf visibility) reduce the reliability of current AI-based approaches.
- Farmers lack an automated, real-time, and expert-validated disease monitoring tool tailored to black pepper cultivation.

Application of Specialization Knowledge & Technologies

Key Domains of Knowledge



IoT & Embedded Systems



Artificial Intelligence & Computer Vision



Software Engineering

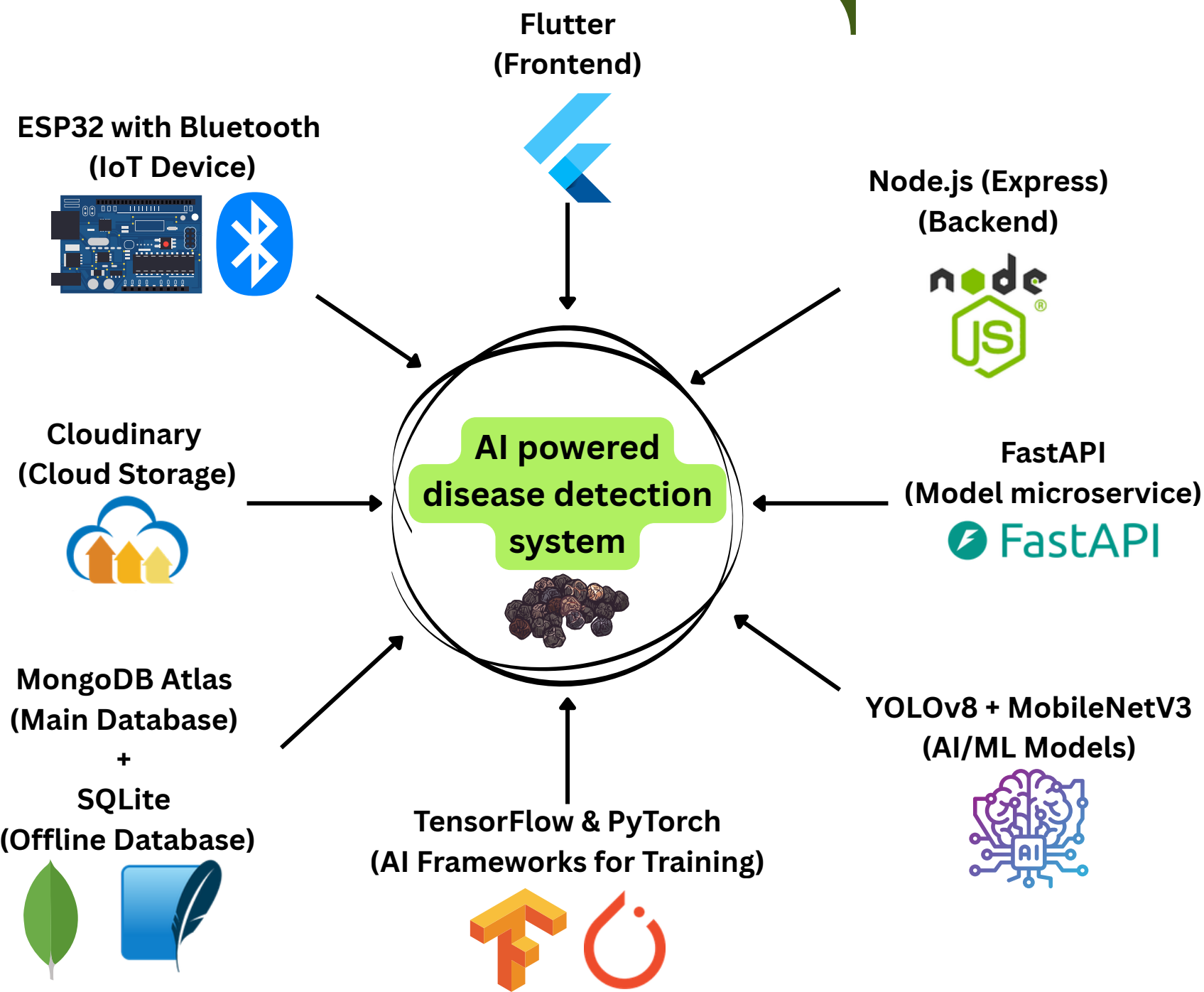


Databases & Cloud Computing



Agricultural Science & Quality Standards

Latest Technologies Utilized

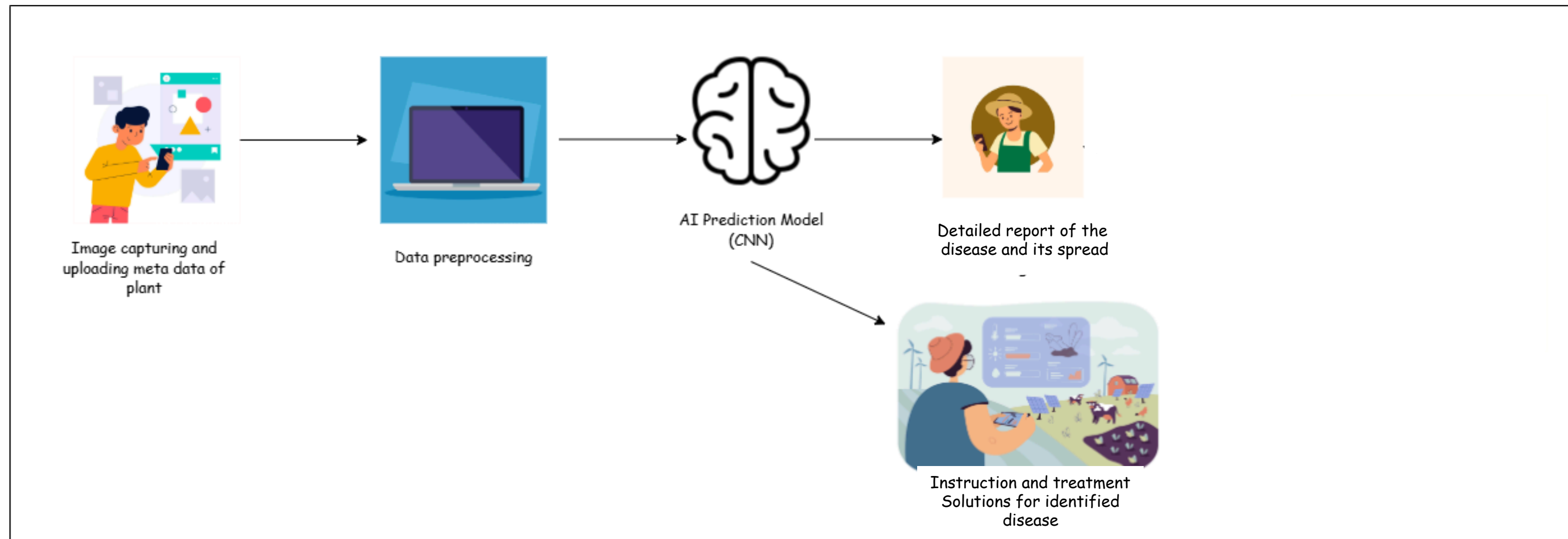


Data availability and ethical clearance

Type of data	source	parameters	volume
Image data of pepper leaves	Field visits to Sri Lankan pepper plantations (selected Kalutara District)	Images of diseased pepper leaves from multiple plants	Dataset of 800-1000 images from 300 pepper plants
Weather and Climate data	OpenWeatherMap API	Daily temperature Humidity Rainfall Sunlight hours	Collecting for 1-2 months per growing cycle
Soil data	IOT sensors	Soil PH Soil moisture Nutrients levels	Collection from pepper plantations

- Farmers’ consent was obtained before capturing images or collecting yield data.
- Data will be anonymized (no farmer names, only plant IDs and GPS coordinates).
- No personal or sensitive data involved (plant-only dataset).
- Datasets will not be published without permission; instead, a research-only dataset will be created.

System Overview

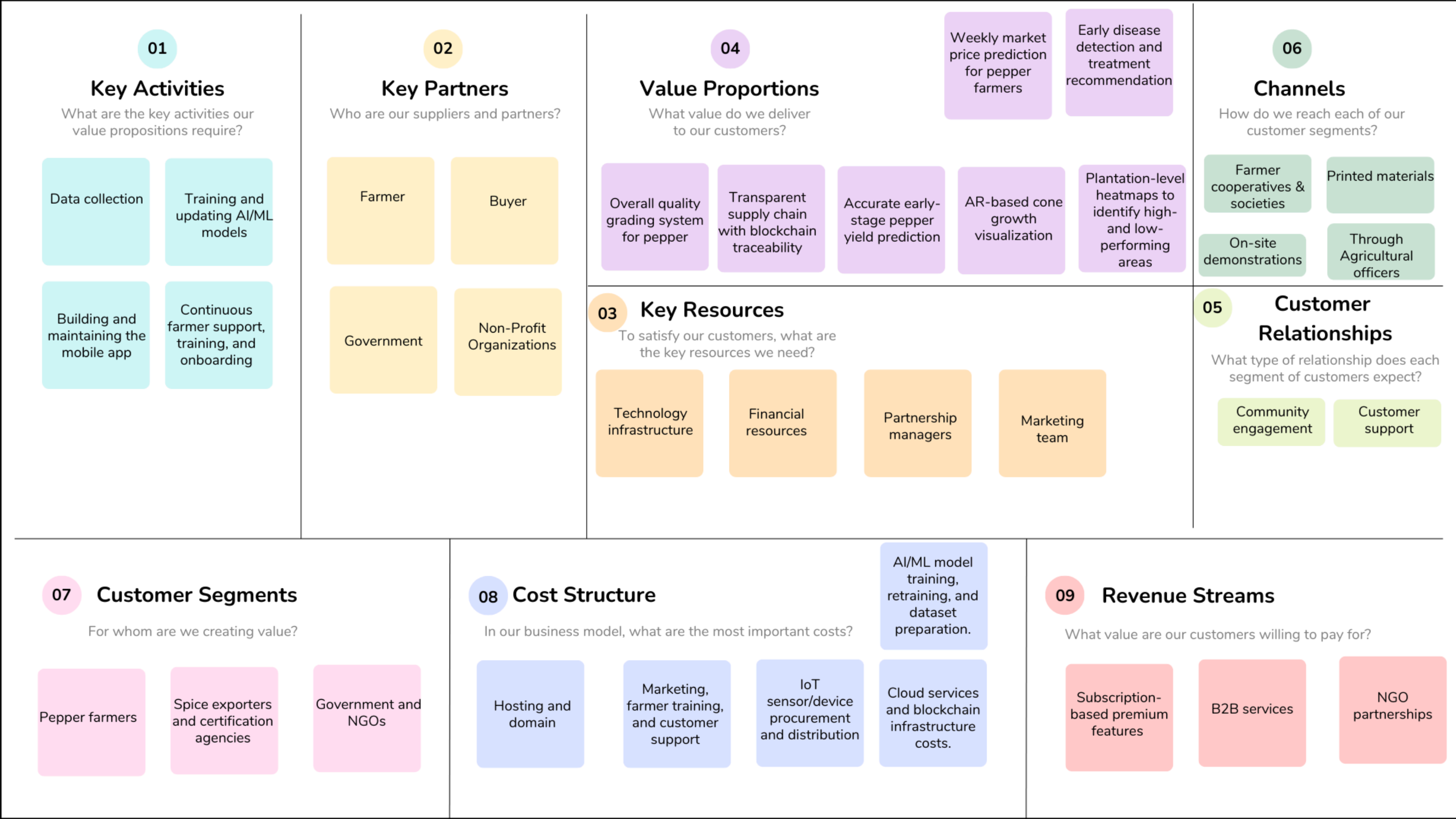


Commercialization

Estimated budget for implementation of the project

Yield Prediction IoT device is used to measure soil pH and moisture levels for accurate yield forecasting.	LKR 18,700
Quality Grading IoT device is used to measure pepper density to ensure standardized grading.	LKR 7040
Disease Detection IoT device is used to identify early signs of diseases.	LKR 7000

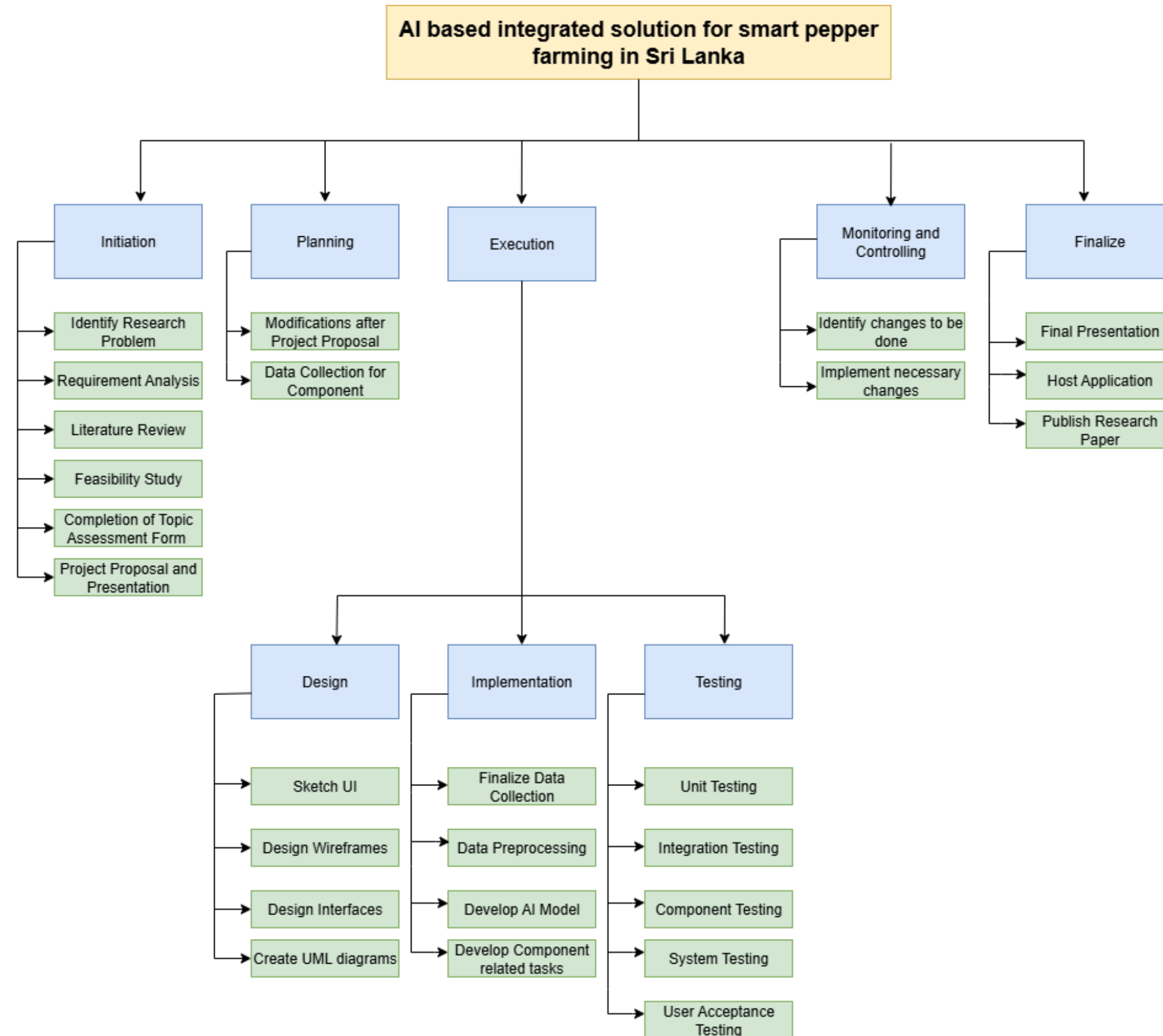
Commercialization



Gantt Chart

Task Name	Progress(%)	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Project Initiation															
Study about research gaps	100														
Deciding appropriate research area	100														
Going through research papers	100														
Finalizing a topic	100														
Requirement Gathering & Analysis															
Identifying functional requirements	100														
Identifying non-functional requirements	100														
Identifying data requirements	100														
Identifying knowledge requirements	100														
Design															
Creating the Figma prototype	0														
Development															
Data preprocessing and model training	0														
Backend Development	0														
Frontend Development	0														
Testing															
Unit and API testing	0														
Integrating Testing	0														
Load testing	0														
User Acceptance Testing	0														
Deployment	0														

Work Breakdown Structure



Functional Requirements

Yield prediction and visualization

- Farmers can capture and upload images of young pepper cones, along with metadata
- Predict cone growth stages directly onto real plants using AR via a smartphone camera.
- Generate interactive heatmaps that visualize yield distribution across the plantation.
- Allow exporting yield forecasts and heatmaps into reports for planning.
- Provide an integrated dashboard

Functional Requirements

Post-Harvest Pepper Quality Grading

- Capture pepper images using the farmer's mobile phone.
- Measure bulk density using the IoT device
- Analyze images with AI models to detect mold, adulteration, berry size, and color.
- Estimate piperine values based on literature and pepper variety.
- Validate GAP certification through the official API.
- Store grading results, images, and certificates securely.
- Generate a digital grading report with quality score and classification.
- Notify buyers automatically if premium-grade pepper is detected.
- Provide farmers with grading feedback and reports via mobile app.

Functional Requirements

Market Price Prediction and Blockchain Integrity

- Provide interpretable AI forecasting for pepper prices.
- Deliver actionable selling recommendations to farmers.
- Record the pepper price and grade of quality securely on a blockchain ledger.
- Enable QR-based traceability for pepper batches.
- Offer unified access for farmers, buyers, and exporters.

AI powered disease detection system

- Uploading or capturing leaf images for disease detection.
- Classify black pepper leaf diseases (healthy vs. disease type)
- Integrate environmental data (temperature, humidity, soil moisture) with image-based predictions.
- Provide visual reports of disease type, spread percentage and recommended actions.

Non-Functional Requirements



SECURITY



AVAILABILITY



PERFORMANCE



USABILITY

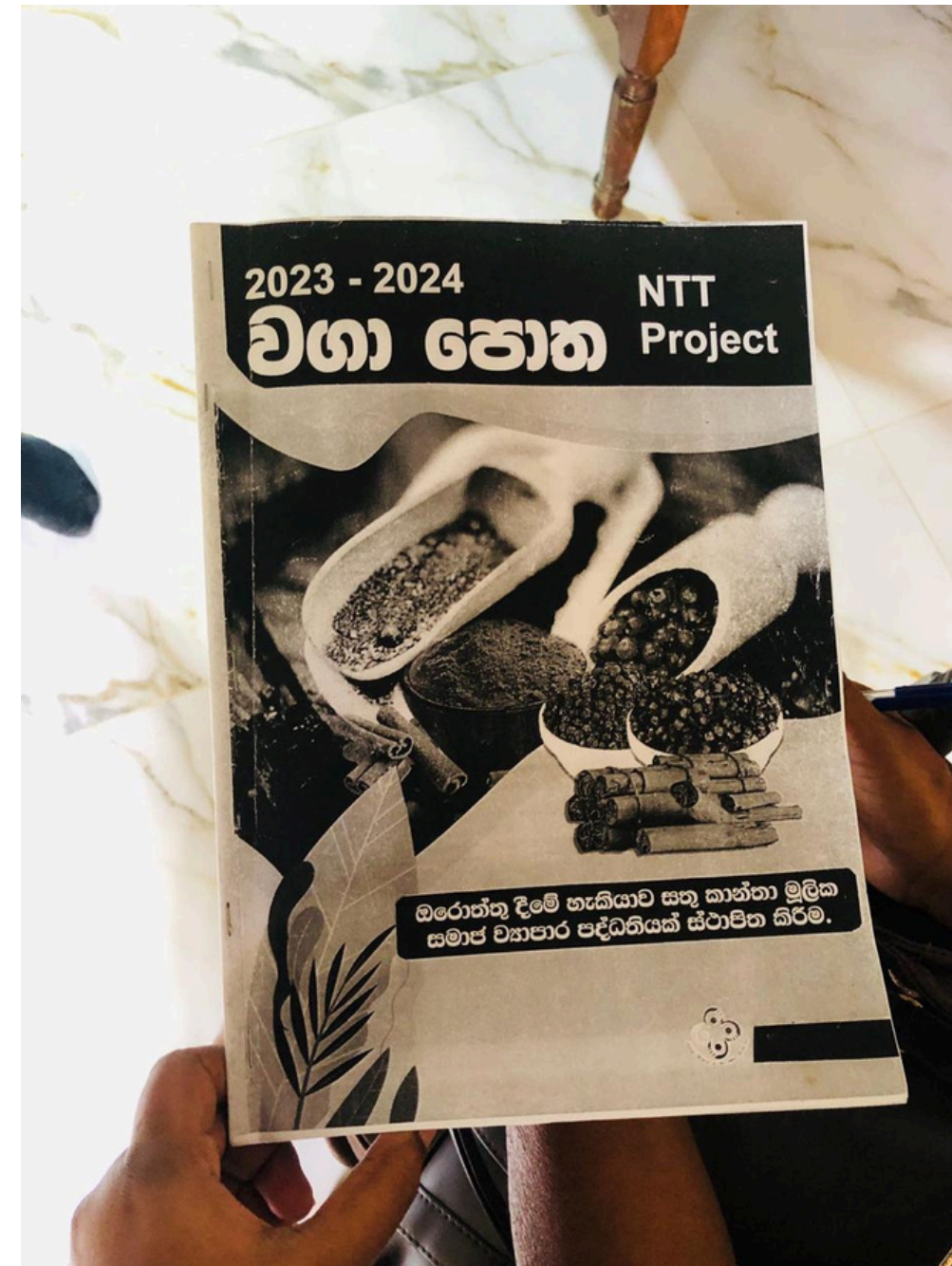


RELIABILITY

Evidences



Evidences



Evidences

06. අස්වැන්න පෙළුම්

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21/05/24	"	-	-	10	6300/-	63000/-	63000/-

04. වගා බිමේ දෛනික සටහන්

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27/05/24	"	07:00	40	1500			25000/-	
28/05/24	"	07:00	40	1500			25000/-	
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Evidences



Thank You

